



CSCE 633-700: Machine Learning

Support Vector Machines

Overview

- Lagrange Optimization
- Support Vector Machines
 - Linearly separable case
 - Non-separable case
 - Hinge Loss
 - Multi-class SVMs

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Lagrange Optimization

Maximize with constraints

$$\begin{aligned} \max_w & f(w) \\ \text{s.t. } & g_i(w) \leq 0, \quad i = 1, \dots, k \\ & h_i(w) = 0, \quad i = 1, \dots, l \end{aligned}$$

1) *Formulate Lagrangian function (primal problem)*

$$L_p(w, \alpha, \beta) = f(w) - \sum_{i=1}^k \alpha_i g_i(w) - \sum_{i=1}^l \beta_i h_i(w)$$

2) *Maximize wrt primal variable w : $\frac{\partial L_p(w, \alpha, \beta)}{\partial w} = 0$*

3) *Substitute the primal variable w and express Lagrangian wrt dual variables α_i, β_i : $L_d(\alpha, \beta)$*

4) *Minimize wrt dual variables and solve for dual variables (dual problem): $\frac{\partial L_d(\alpha, \beta)}{\partial \alpha_i} = 0, \frac{\partial L_d(\alpha, \beta)}{\partial \beta_i} = 0$*

5) *Recover the solution (for the primal variables) from the dual variables*

Lagrange Optimization

Minimize with constraints

$$\begin{aligned} \min_w & f(w) \\ \text{s.t. } & g_i(w) \leq 0, \quad i = 1, \dots, k \\ & h_i(w) = 0, \quad i = 1, \dots, l \end{aligned}$$

1) *Formulate Lagrangian function (primal problem)*

$$L_p(w, \alpha, \beta) = f(w) + \sum_{i=1}^k \alpha_i g_i(w) + \sum_{i=1}^l \beta_i h_i(w)$$

2) *Minimize wrt primal variable w : $\frac{\partial L_p(w, \alpha, \beta)}{\partial w} = 0$*

3) *Substitute the primal variable w and express Lagrangian wrt dual variables α_i, β_i : $L_d(\alpha, \beta)$*

4) *Maximize wrt to dual variables and solve for dual variables (dual problem): $\frac{\partial L_d(\alpha, \beta)}{\partial \alpha_i} = 0, \frac{\partial L_d(\alpha, \beta)}{\partial \beta_i} = 0$*

5) *Recover the solution (for the primal variables) from the dual variables*

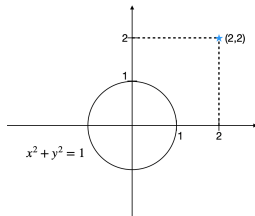
Lagrange Multipliers

Turn a constrained optimization problem into an unconstrained optimization problem by absorbing the constraints into the cost function, weighted by the *Lagrange multipliers*

Example: Find point on the circle $x^2 + y^2 = 1$ closest to the point $(2, 2)$

- Minimize $F(x, y) = (x - 2)^2 + (y - 2)^2$ subject to the constraint $x^2 + y^2 - 1 = 0$.
- Absorb the constraint into the cost function, after multiplying the *Lagrange multiplier* $\alpha > 0$:

$$F(x, y, \alpha) = (x - 2)^2 + (y - 2)^2 + \alpha(x^2 + y^2 - 1).$$



Lagrange Multipliers

Formulate Lagrangian (primal problem):

$$F(x, y, \alpha) = (x - 2)^2 + (y - 2)^2 + \alpha(x^2 + y^2 - 1), \text{ with } \alpha > 0$$

The optimization problem becomes $\max_{\alpha} \min_{x,y} F(x, y, \alpha)$

Minimize $\min_{x,y} F(x, y, \alpha)$

$$\frac{\partial F}{\partial x} = 2(x - 2) + 2\alpha x = 0 \Rightarrow x = \frac{2}{1 + \alpha}$$

$$\frac{\partial F}{\partial y} = 2(y - 2) + 2\alpha y = 0 \Rightarrow y = \frac{2}{1 + \alpha}$$

We substitute x, y in the Lagrangian and express it in terms of its dual form wrt α and maximize it

$$\frac{\partial F}{\partial \alpha} = x^2 + y^2 - 1 = 0 \Rightarrow \left(\frac{2}{1 + \alpha}\right)^2 + \left(\frac{2}{1 + \alpha}\right)^2 = 1 \Rightarrow \alpha = 2\sqrt{2} - 1$$

Recover the solution for the x, y (x, y) = $(1/\sqrt{2}, 1/\sqrt{2})$